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SEMIOTAX: EQUATIONAL MODELS FOR THE LINGUISTIC SENTENCE & THE ‘COMPUTATIONAL THOUGHT’¹

Francisco José López Quintana²

fjlq.cathedra@yahoo.com

Abstract

This paper presents several equational models designed for application in Natural Language Processing. To this end, a formal language for sentence analysis is employed in order to introduce four prototypical models of linguistic syncretism derived from the universal equation of the sentence. Subsequently, an alternative analytical perspective is adopted to explore the different structural configurations of which every sentence is composed. These methodological steps ultimately lead to the formulation of an archetypal pattern referred to as ‘computational thought’.

Key Words: human thought, computational thought, sentence equations, morpho-functional dimension, configurational dimension

1. Introduction

This paper proposes a formal encoding of the sentence within the framework of the theory known as *Sentence Semiotax* (henceforth SS).³ The analysis is developed through the use of a formal language that allows for the systematic modelling of sentence structure and meaning, in accordance with the principles and methodological foundations of SS.

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² He holds a Master’s Degree of Professional Training in Artificial Intelligence and Big Data, a Higher Technician Degree in Computer Science and Communications, and is an AI computer scientist. He also holds a PhD in Philology, a Master’s Degree in English Philology, French Philology, and Romance Philology, as well as a Bachelor’s Degree in Translation and Interpreting (University of Granada). He is a Professor of English Language and has also taught English Literature and French Language at several academic institutions, including the University of Almería, the University of Granada, and UNED (Spain). His research focuses on the intersection of Semiotics, Language, and Computer Science. His principal contribution, developed in collaboration with Dr. Gloria López Ramos, is the formulation of a linguistic theory called *Semiotax*, which centers on the sentence and provides an innovative analysis aimed at constructing a sentence algorithm and its universal equation (López Quintana & López Ramos, 2021).

³ The original Spanish title is *Semiotaxis de la Oración (SO)* (López Quintana & López Ramos, 2021, p. 19).

Moreover, given the impossibility of identifying patterns common to all languages from a strictly formal, namely, exclusively morphosyntactic, perspective, despite the numerous attempts in this direction, particularly within *Generative Grammar* and its various ramifications, it becomes necessary to adopt a broader analytical framework. Specifically, it is possible to integrate the lexical-semantic and morphosyntactic levels within another domain referred to as the morpho-functional level (López Quintana & López Ramos, 2021, p. 47). At this level, the former domains converge to account for the relations established in each sentence between the verb and its arguments, or more precisely, the relations of the verb, as the head of the Vectorial connector (or Vector), in interaction not only with the Referential element (or Referent) but also with the Terminal element (or Terminus) (López Quintana & López Ramos, 2021, pp. 19–20).

It should further be noted that, in order to carry out a morpho-functional analysis of the sentence, it is first necessary to classify the words of a language into two main groups: lexical components and grammatical components. Lexical components are those words that possess the capacity for expansion. Within this group, a further distinction must be made between (a) Nominalisers, which include nouns, pronouns,⁴ and adjectives, and (b) Vectorisers, which are verbs.⁵ Grammatical components, by contrast, lack this expansive capacity and are represented by the remaining word classes. Adverbs may be assigned either to the lexical or to the grammatical group, depending on their functional capacity for expansion (López Quintana & López Ramos, 2021, pp. 247–248).

Consequently, the present study aims to explore the analytical potential of the universal equation of the sentence (3),⁶ offering not only its methodological development at the **morpho-functional level** but also a new dimension that provides a clearer perspective on the dual **configurational nature** of every sentence as a TERNARY STRUCTURE, whose elements can be expanded as BINARY STRUCTURES (López Quintana & López Ramos, 2021, pp. 247, 273–274).

⁴ Pronouns are usually classified as grammatical words; however, for SS, following scholars who regard them as a subclass of nouns (Huddleston & Pullum, 2002, p. 425; Aarts, 2011, p. 29), they are treated as lexical components because they have the capacity for expansion, unlike grammatical components, which lack this property (López Quintana & López Ramos, 2021, pp. 248–249).

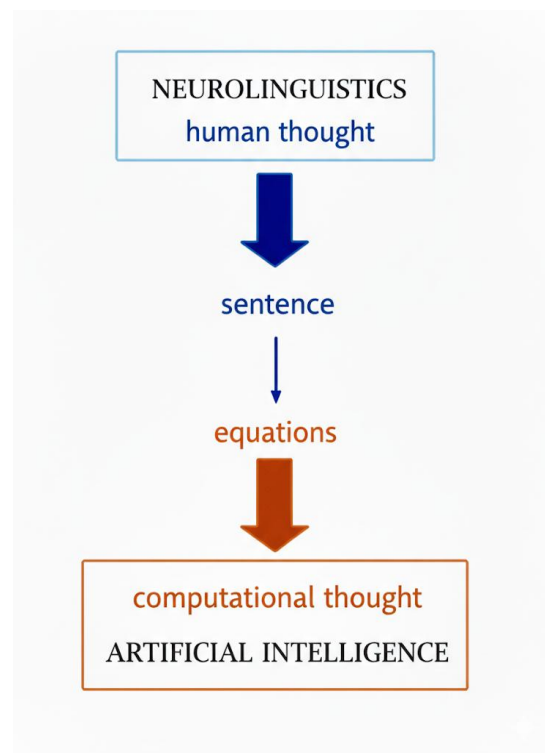
⁵ Verbs in the infinitive form (and, in English, also gerunds) can, in certain cases, function as Nominalisers.

⁶ It is also worth emphasizing that linguistics shares with the human sciences, such as philosophy, an abstract and therefore less precise dimension related to the content level of language. At the same time, it exhibits clear affinities with the experimental sciences, such as physics or chemistry, insofar as its material and thus more concrete dimension concerns the level of expression. This dual nature has enabled the development of mathematical models applied to grammar over recent decades. Nevertheless, it would be a mistake to disregard either of these two aspects, as their interrelation must be taken into account when addressing a field that remains insufficiently explored, namely the formulation of equations such as those presented here. For example, in (i), a purely formal analysis would identify a beginning (*Robert*) and an end (*yesterday*). However, the analysis remains incomplete without a functional perspective, which shows that *the article* constitutes the terminal element at the morpho-functional level where argument-verb relations are established (López Quintana & López Ramos, 2021, pp. 19–20, 77–98). This methodological approach allows for the formulation of a preliminary equation (2), ultimately leading to the universal equation of the sentence (3), also referred to as the equation in the morpho-functional dimension (López Quintana & López Ramos, 2021, p. 246).

(i) *Robert translated the article yesterday.*

Finally, the morpho-functional and configurational dimensions outlined above make it possible to articulate what may be defined as 'computational thought' through a set of equations that constitute a bridge between neurolinguistics and artificial intelligence. By 'computational thought' we refer to an intellectual process, characterised by specific features, which is generated in the human mind ('thought') while simultaneously operating according to an algorithmic logic ('computational'). Although it would be imprudent – despite the arguments advanced by the *Computational Theory of Mind* – to claim, in the absence of definitive empirical evidence, that all human thought functions computationally, there is little doubt that any thought can be expressed algorithmically, regardless of whether it was originally generated in that manner, by means of a sentence. On this basis, we adopt the following postulate: if a thought is defined as the relation established between two ideas, and if a sentence is regarded as the expression of a thought,⁷ then every sentence must consist of three components (an initial element, a final element, and a connector) in order to transmit meaning from one concept to another through a mediating element. In this sense, transitivity emerges as the linguistic property that transforms a sequence of words into a TERNARY STRUCTURE, that is, a sentence (López Quintana & López Ramos, 2021, p. 45), which can be formally represented by precise equations. Thus, we would have: (1).

(1)



⁷ It is important to clarify that this paper does not address the nature of thought, which remains a subject of considerable debate. Rather, because one of our primary concerns is the encoding of ideas, thought must first be transformed into an easily manageable instrument, namely, the sentence, as the sole means of processing it through a formal language. Consequently, the relationship between thought and language is approached here as a methodological necessity, rather than as a matter of theoretical contention.

The detailed morpho-functional analysis shows the distribution of the Referential element, Vectorial connector, Terminal element, and Circumstance element, as well as the internal structure of the subordinate clause. This analysis can be formalised as follows:

$$(4.4) S = \alpha_0.a + \alpha_0 + V_0.a + V_0 + \omega_0 + [\odot]$$



$$[\odot].a + [\odot]\alpha + [\odot]V + [\odot]\omega_0.a + [\odot]\omega_0 + [\odot]\odot_0.a + [\odot]\odot_0.b + [\odot]\odot_0$$

Similarly, complex Vectorial connectors may occur, as illustrated in (5). To recognise the universal equation (3) in such cases, lexicographic evidence – for example, from the *Oxford Advanced Learner's Dictionary of Current English* (OALD) – shows that the phrasal verb *hand out* has the following structure: “hand something out (to somebody),” with two arguments: one compulsory (“something”) and another one optional (“to somebody”), and is defined as “to give a number of things to the members of a group” (Hornby, 2015). Once *yesterday* has been identified as a Circumstance element, for its temporal meaning, and *the company* as the Referential element or Referent (as the morphosyntactic and lexical-semantic References),¹² the groups *the awards* and *to its employees* can be appropriately classified. The first element (*the awards*) characterises the transitivity of the verb (syntactic level) and, thus, it is a compulsory element present in its structure. The second element also has an argument feature,¹³ although this does not imply that it always has to appear explicitly (6), unlike the first one (7). Certainly, a sentence like (7) would be incorrect because of the absence of its Direct Object. If we now use the parameters related to the PRINCIPLE OF TERNARY STRUCTURE underlying every sentence (López Quintana & López Ramos, 2021, p. 45), we will identify the group *to its employees* as the Terminal Element (or Terminus) and *the awards* as a Nuclear supplement included within the Vectorial connector (or Vector), which has *handed out* as its head:¹⁴ (8) and (9).

(5) *The company handed out the awards to its employees yesterday.*

(6) *The company handed out the awards (to its employees) yesterday.*

¹² The morphosyntactic and lexical-semantic References constitute the Referent or Referential element (López Quintana & López Ramos, 2021, pp. 92–94).

¹³ It is highly pertinent to examine the role that dictionaries must play in the linguistic and computational analysis of sentences if a truly scientific study is to be undertaken, i.e., without relying on the speakers' subjectivity in a specific language. However, due to its complexity, this issue will be addressed in a separate article.

¹⁴ It is impossible to explain here the reasons why the Vectorial connector consists of the group *handed out the awards* from the point of view of the SS; for this, see López Quintana & López Ramos (2021, pp. 108–111). Suffice it to say that, in this case, the reason this structure constitutes a morpho-functional unit is its semantic similarity to the verb *awarded*.

(7) * *The company handed out to its employees.*

(8) *The company handed out the awards to its employees yesterday.*

\Vectorial head Nuclear supplement/

Vectorial connector

(9) *The company handed out the awards to its employees yesterday.*

Referential element

Vectorial connector

Terminal element

Circumstance element

α

V

ω

$\odot$

The universal equation in its form (2) enables the formulation of four prototypical equational models of syncretism: an intransitive sentence (10.1), a sentence in which the Referential element is incorporated into the Vectorial connector (11.1), a sentence with an implicit verb (12.1), and an intransitive sentence without an explicit Referential element (13.1). In all cases, the underlying ternary structure persists, although the number of overt lexical units is reduced through incorporation.

- In (10.1) we have the Terminal element combined with the Referential element.

(10.1) *The tree grew.*
 $\alpha = \omega \quad V$

(10.2) $S = \alpha + V + \omega$

(10.3) $\alpha = \omega \longrightarrow \omega = \varepsilon; S = \alpha + V + \varepsilon.$ *The tree grew.*
 $\alpha + V + \varepsilon$

(10.4) $\alpha + \varepsilon = \alpha_\varepsilon; \mathbf{S} = \mathbf{\alpha}_\varepsilon + \mathbf{V}.$ *The tree grew.*
 $\mathbf{\alpha}_\varepsilon \quad \mathbf{V}$

$[\varepsilon: \textit{ensamblaje}]$ ['blending']¹⁵

- Now we will see the formal expression of a sentence where the Referential element (*you*) is not shown as an independent word (11.1) but joined to the Vectorial connector (*close[-Ø]*).

(11.1) *Close the cupboard.*
 $\alpha \subset V \quad \omega$

(11.2) $S = \alpha + V + \omega$

(11.3) $\alpha \subset V \longrightarrow \alpha = \varphi; S = \varphi + V + \omega.$ *Close the cupboard.*
 $\varphi + V + \omega$

(11.4) $\varphi + V = V_\varphi; \mathbf{S} = \mathbf{V}_\varphi + \mathbf{\omega}.$ *Close the cupboard.*
 $\mathbf{V}_\varphi \quad \mathbf{\omega}$

$[\varphi: \textit{fusión}]$ ['merging']

¹⁵ The Greek letters ε , φ , ι , and σ correspond to the initials of the Spanish words *ensamblaje*, *fusión*, *integración*, and *síntesis*.

- Another form of syncretism is found when the Vectorial connector is explicitly absent: (12.1).

(12.1) *Jason, a fool.* [*Jason (is) a fool*]¹⁶

(12.2) $S = \alpha + V + \omega$

(12.3) $V \subset \omega \longrightarrow V = \iota; S = \alpha + \iota + \omega.$ $\frac{\text{Jason, a fool.}}{\alpha + \iota + \omega}$

(12.4) $\iota + \omega = \omega_{\iota}; \mathbf{S} = \alpha + \omega_{\iota}.$ $\frac{\text{Jason, a fool.}}{\alpha \quad \omega_{\iota}}$

[ι : *integración*] ['integration']

- Finally we will show the most intense syncretism that appears when both the Referential element (*you*) and the Terminal one (*you*) are blended and incorporated into the Vectorial connector:¹⁷ (13.1).

(13.1) $\frac{\text{Come!}}{\alpha \subset V; \omega = \alpha}$

(13.2) $S = \alpha + V + \omega$

(13.3) $\alpha \subset V \longrightarrow \alpha = \varphi; \omega = \alpha \longrightarrow \omega = \varepsilon; S = \varphi + V + \varepsilon.$ $\frac{\text{Come!}}{\varphi + V + \varepsilon}$

(13.4) $\varphi + \varepsilon = \sigma; \sigma + V = V_{\sigma}; \mathbf{S} = \mathbf{V}_{\sigma}.$ $\frac{\text{Come!}}{\mathbf{V}_{\sigma}}$

[σ : *síntesis*] ['synthesis']

3. Sentence equation in the configurational dimension: equation of sentence structure

This section is aimed to explore a further dimension of the sentence equation by examining the configurational levels displayed by every sentence, as proposed within SS (López Quintana & López Ramos, 2021, pp. 247, 274–277):

¹⁶ When the Vectorial connector is omitted (usually a copulative verb), it is considered to have been incorporated into the Terminal element, leaving a comma as a meaningful mark of its implicit presence (López Quintana & López Ramos, 2021, pp. 168, 177).

¹⁷ For a detailed explanation of this syncretism, see López Quintana & López Ramos (2021, pp. 121–122).

1st Configuration: it corresponds to the Sentence as such. It is identified whenever a TERNARY STRUCTURE can be recognised, composed of three essential Sentence Segments: the Referential element, the Vectorial connector, and the Terminal element. In addition, optional elements such as Circumstance elements (and Adjunct elements) may also be present.

2nd Configuration: it corresponds to the internal composition of the aforementioned Sentence Segments, whose elements may be expanded by means of a BINARY STRUCTURE.

Moreover, we will now add (to the 1st Configuration) what we term the SINGULAR STRUCTURE, which is associated with the optional Circumstance elements (and Adjunct elements). Accordingly, equation (14) is established as the equation of sentence structure (S_2) or the equation in the configurational dimension:

$$(14) S_2 = \tau + m\gamma + n\beta \quad (m, n \geq 0).$$

Within equation (14), the following elements are defined:

- (a) S_2 denotes equation of sentence structure.
- (b) The symbol τ represents the compulsory TERNARY STRUCTURE and is defined as $\tau = 3l_0$, where l_0 designates the lexical matrix (head) of each essential element of the sentence (referred to as a 'funseme'¹⁸ according to López Quintana & López Ramos, 2021, p. 78), which can be a Referential element, a Vectorial connector, or a Terminal element. When one syncretism occurs, τ is adjusted accordingly: $\tau = 3l_0 - 1$ in cases (10.1), (11.1), and (12.1), and $\tau = 3l_0 - 2$ in case (13.1).¹⁹ Although the ternary structure remains conceptually intact in these cases, the number of overt lexical units is reduced due to incorporation.
- (c) The symbol γ denotes the optional SINGULAR STRUCTURE associated with each Circumstance or Adjunct element. Consequently, $m\gamma$ represents the number of such elements in a sentence and corresponds exactly to the number of their lexical matrix elements (heads). Thus, $m\gamma = ml_{\gamma 0}$, where $l_{\gamma 0}$ is the lexical matrix (head) of each Circumstance element or Adjunct element and $m \geq 0$.
- (d) The symbol β designates the optional BINARY STRUCTURE present within both the TERNARY STRUCTURE and the SINGULAR STRUCTURE. Accordingly, $n\beta$ represents the number of binary structures in a sentence

¹⁸ Hence, the subscript f in $3l_0$; the subscript 0 indicates that it is a matrix (head), as noted above.

¹⁹ In these situations, the ternary structure persists; however, the number of lexical units differs, since one of them incorporates the others into itself.

and coincides with the number of additional (non-matrix or non-nuclear) lexical elements contained within those structures. So, $n\beta = nl_a$, where l_a denotes an additional (non-matrix or non-nuclear) lexical element and $n \geq 0$.

From equation (14), the total number of lexical items in a sentence (S_l) can therefore be expressed as follows:

$$(15) S_l = 3l_0 + ml_{q0} + nl_a \quad (m, n \geq 0).$$

To illustrate the applicability of equation (15), we analyse the following sentence:

(16) *Some Spanish fishermen from the boat run aground on the sand set up tents on the eastern beach at the beginning of July.*

The analysis yields the following results:

(a) $\tau = 3l_0$, corresponding to $\tau = \text{fishermen, set up, and tents}$; there is only one TERNARY STRUCTURE with three matrices (heads) of essential elements: Referential element (*fishermen*), Vectorial connector (*set up*), and Terminal element (*tents*).

(b) $m\gamma = ml_{q0}$, corresponding to $2l_{q0} = \text{beach and beginning}$; $2l_{q0} = 2\gamma$, so we have two SINGULAR STRUCTURES, since we have identified two matrices (heads) corresponding to two Circumstance elements.

(c) $n\beta = nl_a$, corresponding to $6l_a = \text{Spanish, boat, run aground, sand, eastern, and July}$. Therefore, $6l_a = 6\beta$, which indicates that, since there are six additional (non-matrix or non-nuclear) lexical elements, we will consequently have six BINARY STRUCTURES.

In short, (17) is obtained by applying (15) to (14), revealing the structural organisation (ternary, binary, and singular), which gives the final result of eleven lexical items in the sentence. Although the same total number of lexical items could arise from a different distribution of structures, the significance of the equation lies primarily in its capacity to reveal the organisational schema underlying any given sentence sequence.

$$(17) S_2 = \tau + m\gamma + n\beta \longrightarrow S_l = 3l_0 + ml_{q0} + nl_a = 3l_0 + 2l_{q0} + 6l_a = 11 \text{ lexical items.}$$

Within the framework of SS as applied to Natural Language Processing, sentences are encoded in an integer format²⁰ through vectorisation, taking into

²⁰ 'Vectorisation' or 'word embedding'.

account only lexical words,²¹ which convey the core semantic content of the sentence across the two configurational dimensions. The first configuration considers only matrix elements, whereas the second configuration displays the non-matrix lexical elements:

- (a) 1st Configuration: it just considers matrix elements.
- (b) 2nd Configuration: it specifies the sense of the matrix by means of non-matrix lexical elements.

For sentence (16), the following 'vectorisations' are obtained:

V = {at, beach, beginning, boat, eastern, fishermen, from, July, of, on, run aground, sand, set up, some, Spanish, tents, the}.

- (a) 1st Configuration: (0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0).

A value of **1** indicates the presence of matrix elements: *fishermen* (6), *set up* (13), and *tents* (16).

- (b) 2nd Configuration: (0, 1, 1, 1, 1, 0, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 0).

A value of **1** indicates non-matrix lexical items that refine the semantic interpretation of the heads: *beach* (2), *beginning* (3), *boat* (4), *eastern* (5), *July* (8), *run aground* (11), *sand* (12), and *Spanish* (15).

However, it is clear that the descriptive scope of the equation above has certain limitations, so it is essential to present, by means of illustrative graphs, a more detailed analysis of both the organization and the variations that can occur in the above-mentioned structures. For this purpose, we will use the following examples: (18), (19), (20), (21), and (22).

(18) *Sheila's red pen is in the drawer.*

α

(19) *We have lost Sheila's red pen.*

ω

(20) *The army inflicted a humiliating defeat on the enemy.*

V

(21) *An accident took place on the unlit narrow street.*

Y

(22) *The boy delivered a resounding slap to his brother.*

V

²¹ It is especially useful in translation because it provides a global perspective on a sentence's meaning, setting aside structural aspects.

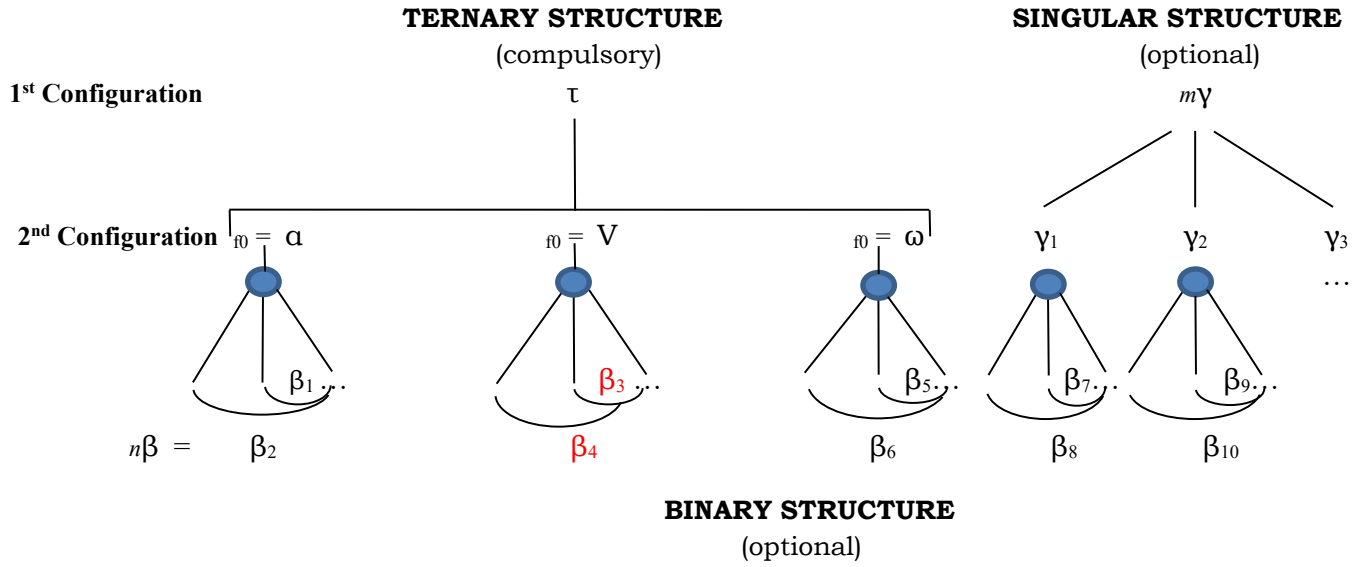


Figure 1: Type 1 of multilevel structure.

(23)

- $\alpha / \omega = \underline{\textit{Sheila}} \textit{ ['s] } \underline{\textit{red}} \underline{\textit{pen.}}$
- β_1 / β_5 *red pen.*
- β_2 / β_6 *Sheila ['s] pen.*
- *Sheila ['s] red pen.*

(24)

- $V = \underline{\textit{inflicted}} \textit{ [a] } \underline{\textit{humiliating}} \underline{\textit{defeat.}}$ ²²
- *inflicted.*
- β_3 *[a] humiliating defeat.*
- β_4 *inflicted [a] humiliating defeat.*

(25)

- $\gamma_1 = \textit{[on the] } \underline{\textit{unlit}} \underline{\textit{narrow}} \underline{\textit{street.}}$
- β_7 *[on the] narrow street.*
- β_8 *[on the] unlit street.*
- *[on the] unlit narrow street.*

²² As noted in footnote 14, we cannot explain here the reasons why *a humiliating defeat* is included in the Vectorial connector together with *inflicted*; for this, see López Quintana & López Ramos (2021, pp. 108–111). At least, the similarity between *inflicted a defeat* and the verb *defeated* can justify considering the former group (comprising the verb and the nominal group) as a functional unit. In the same way, we will see below that the group *delivered a resounding slap* constitutes a Vectorial connector, since it forms a functional unit; here, we can also observe the semantic similarity between *delivered a slap* and the verb *slapped*.

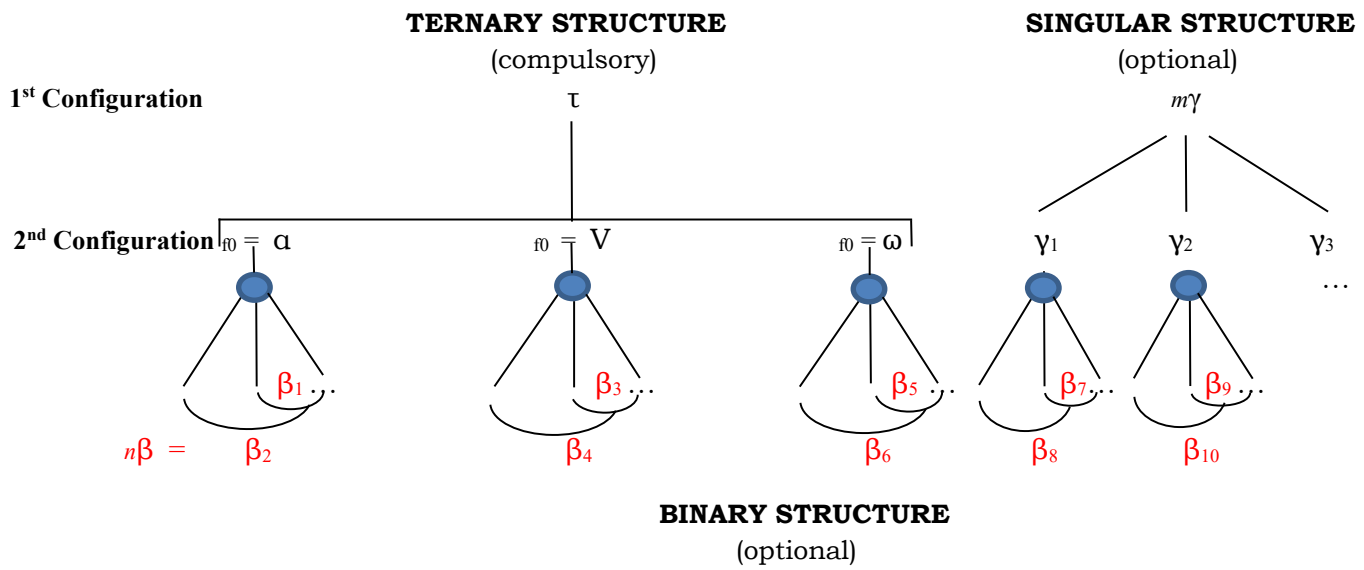


Figure 2: Type 2 of multilevel structure.

(26)

- $\alpha / \omega = \underline{\textit{Sheila}} \textit{ ['s] } \underline{\textit{red}} \textit{ pen.}$
 - $\textit{ Sheila ['s] .}$
 - $\beta_1 / \beta_5) \textit{ red pen.}$
 - $\beta_2 / \beta_6) \textit{ Sheila ['s] red pen.}$

(27)

- $V = \underline{\textit{delivered}} \textit{ [a] } \underline{\textit{resounding}} \textit{ slap.}$
 - $\textit{ delivered.}$
 - $\beta_3) \textit{ [a] resounding slap.}$
 - $\beta_4) \textit{ delivered [a] resounding slap.}$

(28)

- $\gamma_1 = \textit{ [on the] } \underline{\textit{unlit}} \textit{ } \underline{\textit{narrow}} \textit{ } \underline{\textit{street.}}$
 - $\textit{ unlit.}$
 - $\beta_7) \textit{ [on the] narrow street.}$
 - $\beta_8) \textit{ [on the] unlit narrow street.}$

Therefore, we have $\beta \subset \tau$, but also $\beta \subset \gamma$ in both figures.

The crucial issue in the foregoing examples concerns the diverse BINARY STRUCTURES that convey differences in meaning. In this respect, these structures will be interpreted in accordance with the convention established in López Quintana & López Ramos (2021, pp. 256–257). With regard to α / ω , structure (23) (Figure 1) corresponds to a case where there is only one pen which is red and belongs to Sheila, whereas structure (26) (Figure 2) represents a situation in which there are several red pens, but the reference is to the one that belongs to Sheila. A similar contrast can be observed with γ_1 : structure (25) (Figure 1) indicates that there is only one street that is both narrow and unlit, as opposed to (28) (Figure 2), where there are several narrow streets, but the reference is made to the one which is unlit.

Moreover, before determining the BINARY STRUCTURES, it is necessary to distinguish within the Vectorial connector between the verb group (Vectoriser), on the one hand, and the noun and adjective groups (Nominalisers), on the other. Therefore, the structure β of V in (24) (Figure 1) appears in red to indicate that it differs from the rest of the nodes in that group; in fact, its structure is identical to that of (27) (Figure 2). Thus, in (24), we first obtain the binary structure, *[a] humiliating defeat*, formed by the lexical elements of the noun and adjective groups (Nominalisers). Subsequently, this structure is combined with the lexical item in the verb group (Vectoriser), *inflicted*, yielding the phrase *inflicted [a] humiliating defeat*. Likewise, in (27), the binary structure is initially formed by the lexical elements of the noun and adjective groups (Nominalisers), namely *[a] resounding slap*, and, is then combined with the lexical item in the verb group (Vectoriser), *delivered*, resulting in the phrase *delivered [a] resounding slap*.

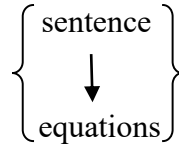
4. Equation of 'computational thought'

On the basis of the preceding equations, it is possible to provide a more precise formulation of the notion of 'computational thought'. This concept may be represented as a transition from human thought, as studied in neurolinguistics, to thought formalised within artificial intelligence by means of sentence-based equations. The relevant equations are as follows:

(29.1)

NEUROLINGUISTICS

Human thought



ARTIFICIAL INTELLIGENCE

Computational thought



$$(29.2) S_1 = mA + 2p + V + n\textcircled{\text{C}} \quad (m, n \geq 0)$$

$$(29.3) S_2 = \tau + m\gamma + n\beta \quad (m, n \geq 0)$$

$$(29.4) \mathbf{K} = \mathbf{2p} + \mathbf{V} + \mathbf{n\gamma} \quad (n \geq 0)^{23}$$

Equation (29.4) defines the structure of 'computational thought', or 'coneme',²⁴ symbolised by K. It consists of three components: two poles (2p) and a Vector (V), taken from the morpho-functional dimension (S₁), and a set of optional elements (nγ), taken from the configurational dimension (S₂). Each of these components (belonging to every 'coneme') constitutes an 'axeme',²⁵ which may be classified as either essential or optional:

- (a) Essential 'axemes': they correspond to the compulsory elements (from S₁) of a 'coneme', namely the poles (represented by 2p), and the Vector, (represented by V), which connects them.²⁶
- (b) Optional 'axemes': these are optional units (from S₂, represented by γ), which integrate Adjunct items and Circumstance elements (at the morpho-functional level of linguistics).

Moreover, every 'axeme' consists of two types of subunits:

- A 'matrix subaxeme', which is a suitable representative of its 'axeme' because it bears the main notional load. It corresponds to nuclear lexical elements in linguistic analysis.
- Optional 'non-matrix subaxemes' are non-essential notional items. They correspond to non-nuclear lexical elements in linguistics.

²³ We must take into account that, in equation (29.2), the symbol A corresponds to Adjunct items and © to Circumstance elements (López Quintana & López Ramos, 2021, p. 246, n. 6). Nevertheless, in (29.4), both are represented by the symbol γ, taken from equation (29.3), so the linguistic distinction (between Adjunct and Circumstance) disappears in the 'computational thought' (29.4). Although in López Quintana & López Ramos (2021, p. 279) it was stated that equation (29.2) could also represent the 'computational thought', it is necessary to make a formal distinction between the linguistic and computational domains, since the latter operates on a more abstract plane than the former and, therefore, must be regarded as different.

²⁴ The term 'coneme' derives from the adjective 'computational' and the noun 'noeme' (Greek origin), meaning 'thought'.

²⁵ 'Axemes' belong to a cognitive-linguistic plane (López Quintana & López Ramos, 2021, pp. 49–52); therefore, these elements are written in double quotation marks (López Quintana & López Ramos, 2021, p. 24).

²⁶ Obviously, the compulsory presence of the poles in a 'coneme' is determined linguistically, i.e., by the semantic level, which conveys the 'computational thought' and, thus, is related to the structural rules of a lexical database (for instance, a dictionary). When we encode the items of a sentence according to a computational model or formula with universal scope, we can convert it into a 'computational thought'.

The formulation of equation (29.4) presupposes the prior application of morpho-functional and configurational analyses through equations (29.2) and (29.3), thereby enabling a transition from linguistic structure to 'computational thought' as we can see below in different languages:

(30) *My injured leg hurts.*

(a) Sentence equation in the morpho-functional dimension:

- $S_0 = \mathbf{mA} + \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{n\textcircled{C}}$; $S_{0(30)} = \mathbf{a} + \mathbf{V} + \mathbf{\omega}$

- $S_{0(30)} = \alpha_{\varepsilon 0} \cdot \mathbf{a} + \alpha_{\varepsilon 1} + \alpha_{\varepsilon 0} + \mathbf{V}_0$

$$\alpha_{\varepsilon 0} \cdot \mathbf{a} = \mathbf{my}$$

$$\alpha_{\varepsilon 1} = \mathbf{injured}$$

$$\alpha_{\varepsilon 0} = \mathbf{leg}$$

$$\mathbf{V}_0 = \mathbf{hurts}$$

$$[\mathbf{S}_1 = \mathbf{mA} + \mathbf{2p} + \mathbf{V} + \mathbf{n\textcircled{C}} \ (m, n \geq 0); S_{1(30)} = 2p + \mathbf{V}]$$

(b) Sentence equation in the configurational dimension:

- $\mathbf{S}_2 = \mathbf{\tau} + \mathbf{m\gamma} + \mathbf{n\beta} \ (m, n \geq 0)$

- $S_{2(30)} = \mathbf{\tau} + \mathbf{\beta}$

$$\mathbf{\tau} = \mathbf{injured \ leg} \ / \ \mathbf{injured \ leg} \ / \ \mathbf{hurts}$$

$$\mathbf{\beta} = \mathbf{injured \ leg}$$

(c) Equation of 'computational thought':

- $\mathbf{K} = \mathbf{2p} + \mathbf{V} + \mathbf{n\gamma} \ (n \geq 0)$

- $K_{(30)} = 2p + \mathbf{V}$

$$p_1 = \text{"[1st person sing.] injured leg"}^{27} \quad \rightarrow \text{'(essential) axeme'}$$

$$p_2 = \text{"[1st person sing.] injured leg"} \quad \rightarrow \text{'(essential) axeme'}$$

$$\text{'matrix subaxeme'} = \text{"[1st person sing.] leg"}$$

$$\text{'non-matrix subaxeme'} = \text{"injured"}$$

$$p_1 = p_2$$

$$\mathbf{V} = \text{"to hurt"}^{28} \quad \rightarrow \text{'(essential) axeme'}$$

²⁷ As mentioned earlier, all the 'axemes' are enclosed in double quotation marks.

²⁸ When 'axemes' refer to verbs, they must appear in the infinitive form (López Quintana & López Ramos, 2021, p. 24).

Although, *prima facie*, lexical elements such as *injured leg* in (30)(b) and 'axemes' in (30)(c), e.g., "[1st person sing.] injured leg," may appear similar, there is a crucial difference. In (30)(b), *injured leg* are lexical words written in italics because they are extracted verbatim from the sentence, leaving aside the grammatical element *my*. In contrast, in (30)(c), "[1st person sing.] injured leg" is enclosed in quotation marks and includes the sense conveyed by the possessive determiner *my* ([1st person sing.]), since it represents an 'axeme' that abstracts all the semantic traits transmitted by the linguistic expression: *my injured leg*, i.e., "speaker's wounded lower limb" (see López Quintana & López Ramos, 2021, pp. 50–51).

(31) *Me duele la pierna herida.* ('My injured leg hurts.')

(a) Sentence equation in the morpho-functional dimension:

- $S_0 = \mathbf{mA} + \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{n\odot}$; $S_{0(31)} = \mathbf{a} + \mathbf{V} + \mathbf{\omega}$

- $S_{0(31)} = \omega_0 + V_0 + \alpha_0.a + \alpha_0 + \alpha_1$

$$\omega_0 = me \text{ ('me')}$$

$$V_0 = duele \text{ ('hurts')}$$

$$\alpha_0.a = la \text{ ('the')}$$

$$\alpha_0 = pierna \text{ ('leg')}$$

$$\alpha_1 = herida \text{ ('injured')}$$

$$[S_1 = \mathbf{mA} + \mathbf{2p} + \mathbf{V} + \mathbf{n\odot} \text{ (} m, n \geq 0 \text{)}; S_{1(31)} = 2p + V]$$

(b) Sentence equation in the configurational dimension:

- $S_2 = \mathbf{\tau} + \mathbf{m\gamma} + \mathbf{n\beta}$ ($m, n \geq 0$)

- $S_{2(31)} = \mathbf{\tau} + \mathbf{\beta}$

$$\tau = me \text{ ('me')} / duele \text{ ('hurts')} / pierna \text{ herida} \text{ ('injured leg')}$$

$$\beta = \underbrace{pierna \text{ herida}} \text{ ('injured leg')}$$

(c) Equation of 'computational thought':

- $\mathbf{K} = \mathbf{2p} + \mathbf{V} + \mathbf{n\gamma}$ ($n \geq 0$)

- $K_{(31)} = 2p + V$

$$p_1 = \text{"pierna herida"} \text{ ('injured leg')} \rightarrow \text{'(essential) axeme'}$$

$$\text{'matrix subaxeme'} = \text{"pierna"} \text{ ('leg')}$$

$$\text{'non-matrix subaxeme'} = \text{"herida"} \text{ ('injured')}$$

$$\begin{aligned}
 p_2 &= \text{"yo"} \text{ ('I')}^{29} && \rightarrow \text{'(essential) axeme'} \\
 p_1 &\neq p_2 \\
 V &= \text{"doler"} \text{ ('to hurt')} && \rightarrow \text{'(essential) axeme'}
 \end{aligned}$$

(32) *J'ai mal à la jambe blessée.* ('My injured leg hurts.')

(a) Sentence equation in the morpho-functional dimension:

- $S_0 = \mathbf{mA} + \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{n\textcircled{C}}$; $S_{0(32)} = \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{\textcircled{C}}$
- $S_{0(32)} = \mathbf{a}_0 + \mathbf{V}_0 + \mathbf{\omega}_0 + \mathbf{\textcircled{C}_0.a} + \mathbf{\textcircled{C}_0.b} + \mathbf{\textcircled{C}_0} + \mathbf{\textcircled{C}_1}$

$$\begin{aligned}
 \mathbf{a}_0 &= j(e) \text{ ('I')} \\
 \mathbf{V}_0 &= ai \text{ ('have')} \\
 \mathbf{\omega}_0 &= mal \text{ ('pain')} \\
 \mathbf{\textcircled{C}_0.a} &= à \text{ ('in')} \\
 \mathbf{\textcircled{C}_0.b} &= la \text{ ('the')} \\
 \mathbf{\textcircled{C}_0} &= jambe \text{ ('leg')} \\
 \mathbf{\textcircled{C}_1} &= blessée \text{ ('injured')} \\
 [\mathbf{S_1} &= \mathbf{mA} + \mathbf{2p} + \mathbf{V} + \mathbf{n\textcircled{C}} \text{ (} m, n \geq 0 \text{)}; S_{1(32)} = 2p + V + \textcircled{C}]
 \end{aligned}$$

(b) Sentence equation in the configurational dimension:

- $S_2 = \mathbf{\tau} + \mathbf{m\gamma} + \mathbf{n\beta}$ ($m, n \geq 0$)
- $S_{2(32)} = \mathbf{\tau} + \mathbf{\gamma} + \mathbf{\beta}$

$$\begin{aligned}
 \mathbf{\tau} &= j(e) \text{ ('I')} / ai \text{ ('have')} / mal \text{ ('pain')} \\
 \mathbf{\gamma} &= jambe blessée \text{ ('injured leg')} \\
 \mathbf{\beta} &= \underbrace{jambe blessée} \text{ ('injured leg')}
 \end{aligned}$$

(c) Equation of 'computational thought':

- $\mathbf{K} = \mathbf{2p} + \mathbf{V} + \mathbf{n\gamma}$ ($n \geq 0$)
- $K_{(32)} = 2p + V + \gamma$

$$p_I = \text{"j(e)"} \text{ ('I')} \quad \rightarrow \text{'(essential) axeme'}$$

²⁹ When 'axemes' refer to pronouns, they are written in the nominative case as a symbol of their abstract nature. Therefore, the pronoun appears as "yo" ('I') instead of "me" ('me'), which is the actual form used in the sentence.

$p_2 = \text{"mal"} \text{ ('pain')}$ $\rightarrow$ '(essential) axeme'

$p_1 \neq p_2$

$V = \text{"avoir"} \text{ ('to have')}$ $\rightarrow$ '(essential) axeme'

$\gamma = \text{"jambe blessée"} \text{ ('injured leg')}$ $\rightarrow$ '(optional) axeme'

'matrix subaxeme' = "jambe" ('leg')

'non-matrix subaxeme' = "blessée" ('injured')

Let us now compare the following examples in order to illustrate the range of processing economy, not only at the level of the linguistic sentence but also in terms of 'computational thought', showing the number of elements resulting from the application of each equation:

(33) *Il a mal à la jambe blessée.* ('His injured leg hurts.')

(1) Sentence equation in the morpho-functional dimension:

➤ $S_0 = \mathbf{mA} + \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{n\textcircled{C}}$; $S_{0(33)} = \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{\textcircled{C}}$

➤ $S_{0(33)} = \mathbf{a}_0 + \mathbf{V}_0 + \mathbf{\omega}_0 + \mathbf{\textcircled{C}_0.a} + \mathbf{\textcircled{C}_0.b} + \mathbf{\textcircled{C}_0} + \mathbf{\textcircled{C}_1} \rightarrow$ **7 ELEMENTS**

$\mathbf{a}_0 = \textit{il}$ ('he')

$\mathbf{V}_0 = \textit{a}$ ('has')

$\mathbf{\omega}_0 = \textit{mal}$ ('pain')

$\mathbf{\textcircled{C}_0.a} = \textit{\grave{a}}$ ('in')

$\mathbf{\textcircled{C}_0.b} = \textit{la}$ ('the')

$\mathbf{\textcircled{C}_0} = \textit{jambe}$ ('leg')

$\mathbf{\textcircled{C}_1} = \textit{blessée}$ ('injured')

$[\mathbf{S_1} = \mathbf{mA} + \mathbf{2p} + \mathbf{V} + \mathbf{n\textcircled{C}} \text{ (} m, n \geq 0 \text{)}; S_{1(33)} = 2p + V + \textcircled{C}]$

(2) Equation of 'computational thought':

➤ $\mathbf{K} = \mathbf{2p} + \mathbf{V} + \mathbf{n\gamma} \text{ (} n \geq 0 \text{)}$

➤ $K_{(33)} = 2p + V + \gamma \rightarrow$ **4 ELEMENTS**

$p_1 = \text{"il"} \text{ ('he')}$ $\rightarrow$ '(essential) axeme'

$p_2 = \text{"mal"} \text{ ('pain')}$ $\rightarrow$ '(essential) axeme'

$p_1 \neq p_2$

$V = \text{"avoir"} \text{ ('to have')}$ $\rightarrow$ '(essential) axeme'

$\gamma = \text{"jambe blessée"} \text{ ('injured leg')}$ $\rightarrow$ '(optional) axeme'

'matrix subaxeme' = "jambe" ('leg')

'non-matrix subaxeme' = "blessée" ('injured')

(34) *His injured leg causes him pain.*

(1) Sentence equation in the morpho-functional dimension:

$$\triangleright \mathbf{S}_0 = \mathbf{mA} + \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{n\odot}; \quad \mathbf{S}_{0(34)} = \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{\odot}$$

$$\triangleright \mathbf{S}_{0(34)} = \alpha_0.a + \alpha_1 + \alpha_0 + \mathbf{V}_0 + \odot_0 + \omega_0 \quad \rightarrow \quad \mathbf{6 \text{ ELEMENTS}}$$

$$\alpha_0.a = \textit{his}$$

$$\alpha_1 = \textit{injured}$$

$$\alpha_0 = \textit{leg}$$

$$\mathbf{V}_0 = \textit{causes}$$

$$\odot_0 = \textit{him}^{30}$$

$$\omega_0 = \textit{pain}$$

$$[\mathbf{S}_1 = \mathbf{mA} + \mathbf{2p} + \mathbf{V} + \mathbf{n\odot} \ (m, n \geq 0); \quad \mathbf{S}_{1(34)} = \mathbf{2p} + \mathbf{V} + \mathbf{\odot}]$$

(2) Equation of 'computational thought':

$$\triangleright \mathbf{K} = \mathbf{2p} + \mathbf{V} + \mathbf{ny} \ (n \geq 0)$$

$$\triangleright \mathbf{K}_{(34)} = \mathbf{2p} + \mathbf{V} + \mathbf{y} \quad \rightarrow \quad \mathbf{4 \text{ ELEMENTS}}$$

$$p_1 = \text{"[3rd person sing.] injured leg"} \quad \rightarrow \text{'(essential) axeme'}$$

$$\text{'matrix subaxeme'} = \text{"[3rd person sing.] leg"}$$

$$\text{'non-matrix subaxeme'} = \text{"injured"}$$

$$p_2 = \text{"pain"} \quad \rightarrow \text{'(essential) axeme'}$$

$$p_1 \neq p_2$$

$$\mathbf{V} = \text{"to cause"} \quad \rightarrow \text{'(essential) axeme'}$$

$$\mathbf{y} = \text{"he"}^{31} \quad \rightarrow \text{'(optional) axeme'}$$

³⁰ The pronoun *him* is a non-argument item (referring to the person in question) and, therefore, is regarded as a Circumstance element in *SS* (López Quintana & López Ramos, 2021, p. 115, n. 145).

³¹ As stated before, when 'axemes' refer to pronouns, they are written in the nominative case to reflect their abstract nature. Accordingly, this pronoun appears as "he" rather than "him", which is the form attested in the sentence.

(35) *Le duele la pierna herida.* ('His injured leg hurts.')

(1) Sentence equation in the morpho-functional dimension:

$$\begin{aligned}
 &\triangleright \mathbf{S}_0 = \mathbf{mA} + \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{n}\textcircled{\text{C}}; \quad \mathbf{S}_{0(35)} = \mathbf{a} + \mathbf{V} + \mathbf{\omega} \\
 &\triangleright \mathbf{S}_{0(35)} = \omega_0 + \mathbf{V}_0 + \alpha_0.a + \alpha_0 + \alpha_1 \quad \rightarrow \quad \mathbf{5 \text{ ELEMENTS}} \\
 &\quad \omega_0 = le \text{ ('him')} \\
 &\quad \mathbf{V}_0 = duele \text{ ('hurts')} \\
 &\quad \alpha_0.a = la \text{ ('the')} \\
 &\quad \alpha_0 = pierna \text{ ('leg')} \\
 &\quad \alpha_1 = herida \text{ ('injured')} \\
 &\quad [\mathbf{S}_1 = \mathbf{mA} + \mathbf{2p} + \mathbf{V} + \mathbf{n}\textcircled{\text{C}} \text{ } (m, n \geq 0); \quad \mathbf{S}_{1(35)} = 2p + \mathbf{V}]
 \end{aligned}$$

(2) Equation of 'computational thought':

$$\begin{aligned}
 &\triangleright \mathbf{K} = \mathbf{2p} + \mathbf{V} + \mathbf{ny} \text{ } (n \geq 0) \\
 &\triangleright \mathbf{K}_{(35)} = 2p + \mathbf{V} \quad \rightarrow \quad \mathbf{3 \text{ ELEMENTS}} \\
 &\quad p_1 = \text{"pierna herida"} \text{ ('injured leg')} \quad \rightarrow \text{'(essential) axeme'} \\
 &\quad \text{'matrix subaxeme'} = \text{"pierna"} \text{ ('leg')} \\
 &\quad \text{'non-matrix subaxeme'} = \text{"herida"} \text{ ('injured')} \\
 &\quad p_2 = \text{"él"} \text{ ('he')}^{32} \quad \rightarrow \text{'(essential) axeme'} \\
 &\quad p_1 \neq p_2 \\
 &\quad \mathbf{V} = \text{"doler"} \text{ ('to hurt')} \quad \rightarrow \text{'(essential) axeme'}
 \end{aligned}$$

(36) *His injured leg hurts.*

(1) Sentence equation in the morpho-functional dimension:

$$\begin{aligned}
 &\triangleright \mathbf{S}_0 = \mathbf{mA} + \mathbf{a} + \mathbf{V} + \mathbf{\omega} + \mathbf{n}\textcircled{\text{C}}; \quad \mathbf{S}_{0(36)} = \mathbf{a} + \mathbf{V} + \mathbf{\omega} \\
 &\triangleright \mathbf{S}_{0(36)} = \alpha_{\varepsilon 0}.a + \alpha_{\varepsilon 1} + \alpha_{\varepsilon 0} + \mathbf{V}_0 \quad \rightarrow \quad \mathbf{4 \text{ ELEMENTS}} \\
 &\quad \alpha_{\varepsilon 0}.a = his \\
 &\quad \alpha_{\varepsilon 1} = injured \\
 &\quad \alpha_{\varepsilon 0} = leg
 \end{aligned}$$

³² Again, the pronoun *le* ('him') in equation (1) is represented in the nominative case as "él" ('he') in equation (2) to reflect the abstract nature of 'axemes'.

$$V_0 = \text{hurts}$$

$$[S_1 = \mathbf{mA} + \mathbf{2p} + \mathbf{V} + \mathbf{n\odot} \ (m, n \geq 0); S_{1(36)} = 2p + V]$$

(2) Equation of 'computational thought':

$$\triangleright \mathbf{K} = \mathbf{2p} + \mathbf{V} + \mathbf{n\gamma} \ (n \geq 0)$$

$$\triangleright K_{(36)} = 2p + V \quad \rightarrow \quad \mathbf{2 \ ELEMENTS}$$

$$p_1 = \text{"[3rd person sing.] injured leg"} \quad \rightarrow \text{'(essential) axeme'}$$

$$p_2 = \text{"[3rd person sing.] injured leg"} \quad \rightarrow \text{'(essential) axeme'}$$

$$\text{'matrix subaxeme'} = \text{"[3rd person sing.] leg"}$$

$$\text{'non-matrix subaxeme'} = \text{"injured"}$$

$$p_1 = p_2$$

$$V = \text{"to hurt"} \quad \rightarrow \text{'(essential) axeme'}$$

If we compare (35) and (36), we can say that the 'coneme' (36) (2), with notions p_1 and p_2 : "wounded lower limb of [3rd person sing.]" and V: "to hurt", is able to be expressed in English but not in Spanish, because in the latter there is a different computational structure to convey this thought: (35) (2). In this regard, comparative examples demonstrate that different linguistic structures may project distinct computational configurations, even when expressing closely related meanings. Consequently, every 'computational thought' can be defined as the sequence of notions modelled according to equation (29.4), which may be realised through either natural or artificial languages. Such 'conemes' may be obtained empirically, by abstraction from linguistic data, or theoretically, by constructing computational models and testing their linguistic realisability.

5. Conclusion

This article has presented a formal framework for the analysis of the linguistic sentence, grounded in the theoretical principles of *Sentence Semiotax*. By articulating morpho-functional, configurational, and computational dimensions, the study has demonstrated that sentence structure can be modelled through a coherent system of equations that preserve linguistic specificity while enabling cross-linguistic abstraction.

In this way, the analyses presented above have been developed through a multilevel, top-down methodological process, which clearly differentiates *Sentence Semiotax* from the notion of 'Merge' proposed in Chomsky's *Minimalist Program*. The latter is characterised by an exclusively binary, bottom-up derivational trajectory (Chomsky, 2007, p. 4). By contrast, *Sentence Semiotax* adopts an onomasiological perspective (Pottier, 1987) on sentence production (Garrett, 1975, 1989), thereby exhibiting a top-down orientation.

This orientation more accurately reflects the cognitive process involved in the elaboration of thought than a purely bottom-up approach. It is precisely this theoretical framework that has enabled the formulation of:

- (a) the **universal sentence equation at the morpho-functional level**, together with its four prototypical models of linguistic syncretism, shown to capture the essential ternary organisation underlying all sentences, independently of their surface realisation;
- (b) the **equation of the sentence structure in the configurational dimension**, which further reveals how this ternary core expands through singular and binary structures, thereby providing an account of internal sentence organisation; and
- (c) the **equation of 'computational thought'**, which offers an abstract framework of thought capable of being algorithmically expressed through language. By introducing the notions of 'coneme', 'axeme', and 'subaxeme', the model establishes a formal interface between linguistic analysis, cognitive representation, and artificial intelligence.

Rather than advancing an ontological claim about the nature of all cognition, the proposed framework provides a methodological tool for encoding thought in a form suitable for computational processing. As such, it opens new avenues for the development of explainable and linguistically grounded NLP systems. Therefore, this framework may be extended by applying the equations presented here to large-scale linguistic corpora, integrating them into computational architectures, or exploring their relevance to language acquisition and cognitive modelling. In this respect, *Sentence Semiotax* may contribute not only to theoretical linguistics but also to the design of formally interpretable models of human and artificial cognition.

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